

# EC2 Deployment Commands & Reference Guide

This document lists every command used to configure your AWS EC2 instance (54.197.10.99), build the application, upload it, and start both the **Client** and **Server** containers.

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## 1. Local Machine Commands (Build & Package)

To prevent memory crashes on the EC2 instance, we compile the React client locally on the Mac development machine and package the build outputs.

### Set SSH Private Key Permissions

```
chmod 400 cryptovault.pem
```

### Compile the Client App Locally

Run inside the client/ directory:

```
npm run build
```

*This generates the compiled static files inside client/dist/.*

### Compress and Package the Repository

Run from the root CryptoVault/ directory:

```
tar -czf archive.tar.gz --exclude="node_modules" --exclude=".git" --exclude="archive.tar.gz" -C . .
```

*This archives client, server, docker configs, and the pre-built dist/ directory, while excluding heavy dependency folders.*

### Upload the Archive to EC2

```
scp -i cryptovault.pem archive.tar.gz ubuntu@54.197.10.99:/home/ubuntu/
```

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## 2. Remote EC2 Configuration Commands

Run these commands on the EC2 instance.

### SSH into the Instance

```
ssh -i cryptovault.pem ubuntu@54.197.10.99
```

### Clean Broken Repos & Install Docker

Run these to cleanly install the official Docker Engine and Docker Compose plugins:

```
# Remove any corrupted docker list entries
sudo rm -f /etc/apt/sources.list.d/docker.list

# Run the official Docker installation script
curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh

# Add the 'ubuntu' user to the docker group so you don't have to type sudo for docker commands
sudo usermod -aG docker ubuntu
rm -f get-docker.sh
```

*Note: Exit the SSH session (exit) and log back in to apply the group changes.*

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### 3. Deploy and Run the Containers on EC2

Run these commands inside the EC2 shell after logging back in.

#### Extract the Code

```
mkdir -p ~/CryptoVault
tar -xzf ~/archive.tar.gz -C ~/CryptoVault
rm ~/archive.tar.gz
```

#### Build and Launch the Containers

```
cd ~/CryptoVault
docker compose -f docker-compose.prod.yml up --build -d
```

#### Verify Container Status and Logs

```
# List all running containers
docker ps

# Check the API server logs
docker logs cryptovault-prod-api
```